DEPARTMENT OF MECHANICAL ENGINEERING

1D Premixed Methane-Air Combustion Simulation

Numerical Study using Detailed Chemical Kinetics (Cantera & GRI-Mech 3.0)

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Tools: Python 3.10 • Cantera 3.2 • GRI-Mech 3.0

Problem Definition & Governing Equations

Physical Scope:

- Planar, 1D freely-propagating laminar flame.
- Inflow: T_u = 300 K, P = 1 atm (101325 Pa).
- Dry Air mixture: O_2 : N_2 = 1 : 3.76.
- Domain width: L = 30 mm.
- Dense sweep: ϕ = 0.65 to 1.35 (15 points).

Primary Objectives:

- Compute laminar flame speed S_L as an eigenvalue.
- Determine peak and adiabatic temperatures.
- Resolve internal flame preheat vs reaction zones.

1. Continuity (Mass Conservation) :

$$\frac{d(\rho u)}{dx} = 0 \quad \rightarrow \quad \dot{m} = \rho u = \text{constant}$$

2. Species Conservation (k = 1, … , 53) :

$$\dot{m}\frac{dY_k}{dx} = -\frac{dj_k}{dx} + \dot{\omega}_k W_k$$

3. Energy Conservation :

$$\dot{m}c_p\frac{dT}{dx} = \frac{d}{dx}\left(\lambda\frac{dT}{dx}\right) - \sum_{k=1}^{K} j_k c_{p,k}\frac{dT}{dx} - \sum_{k=1}^{K} h_k \dot{\omega}_k W_k$$

4. Thermal Equation of State (Ideal Gas Law) :

$$P = \rho R_{\text{spec}} T = \rho\left(\frac{R_u}{\bar{W}}\right)T$$

Chemical Mechanism & Adaptive Numerical Grid

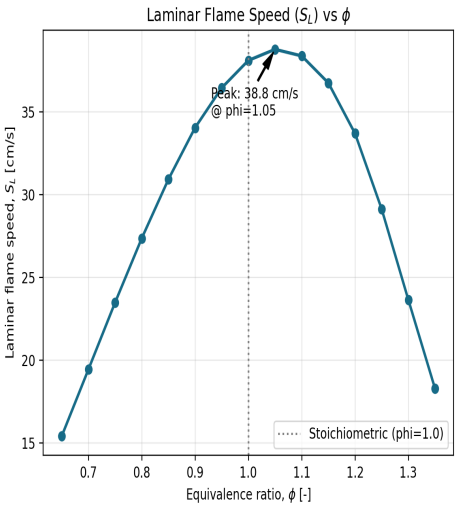
Detailed Kinetic Mechanism (GRI-Mech 3.0):

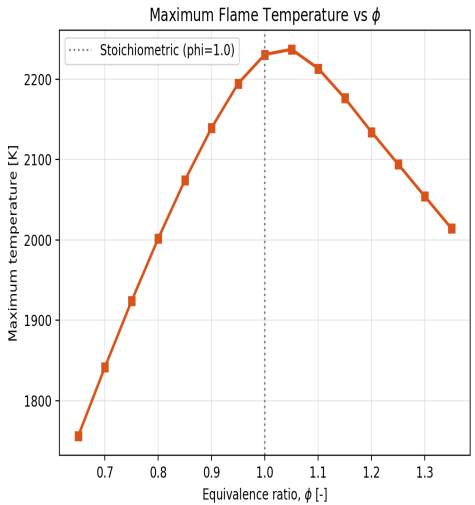
- **GRI:** *Gas Research Institute* (GRI-Mech 3.0 benchmark).
- **53 chemical species**, **325 elementary reactions**.
- Industry standard for natural gas and methane combustion modeling.
- Includes comprehensive NO_x intermediate pathways.
- NASA polynomials for thermodynamic state (H, S, c_p).
- Mixture-averaged transport formulation (λ, μ, D_{km}).

Cantera Numerical Methodology:

- **Boundary Value Problem (BVP):** Solved using a hybrid damped Newton-Raphson iteration.
- **Eigenvalue Problem:** Inlet velocity $u(0) = S_L$ is solved to balance flame propagation.
- **Adaptive Mesh Refinement:**
  - Initial uniform grid: 16 points.
  - Refinement criteria: slope=0.05, curve=0.08.
  - Final converged mesh: **196 to 240 points** concentrated strictly within the reaction zone.

Results: Flame Speed & Temperature vs Mixture Strength

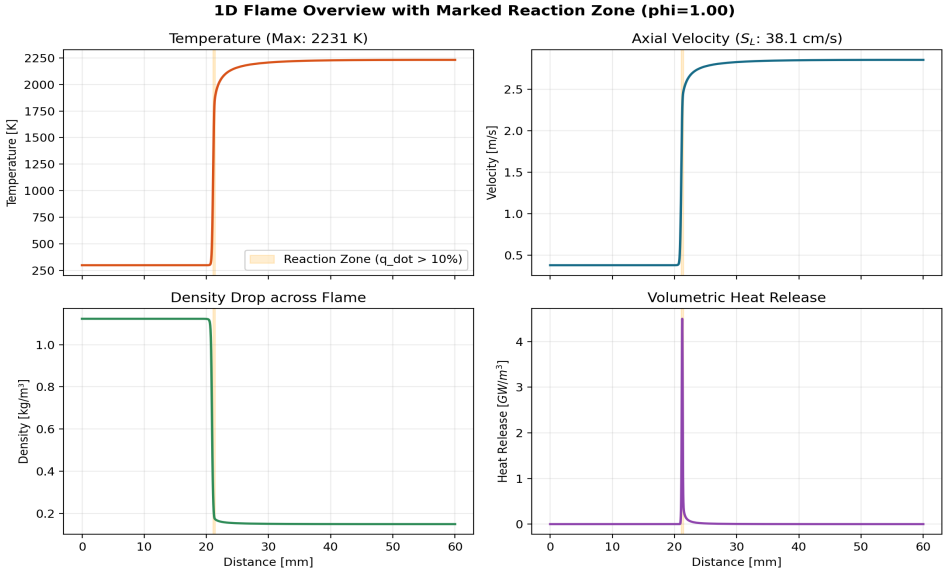




ϕ	Regime	S_L [cm/s]	T_max [K]
0.65	Lean Limit	15.41	1755.8
0.80	Lean	27.35	2001.6
0.90	Lean	34.03	2139.3
1.00	Stoichiometric	38.09	2230.7
1.05	Peak Speed & T	38.76	2237.1
1.10	Slightly Rich	38.36	2213.4
1.20	Rich	33.68	2134.3
1.35	Rich Limit	18.28	2014.3

- **Dense 15-point sweep ($\Delta\phi$ = 0.05)** yields a continuous smooth curve.
- **Global peak occurs at ϕ = 1.05** (S_L = 38.76 cm/s, T_max = 2237.1 K).

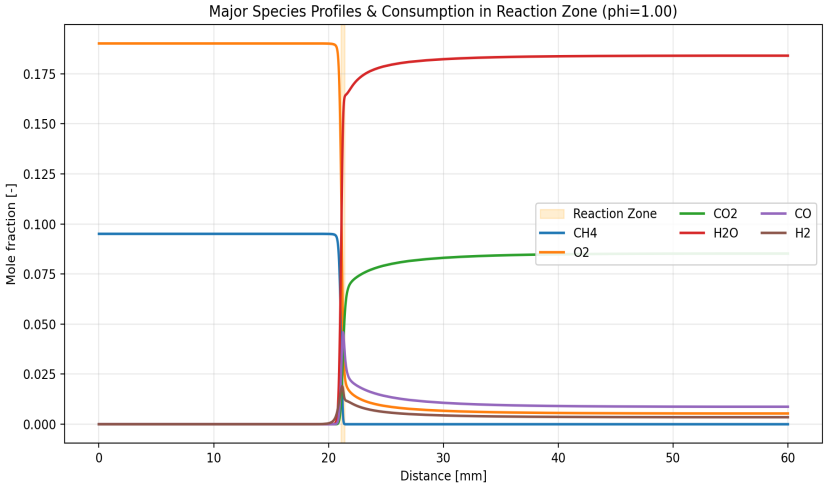
Results: 1D Flame Physical Structure (ϕ = 1.00)

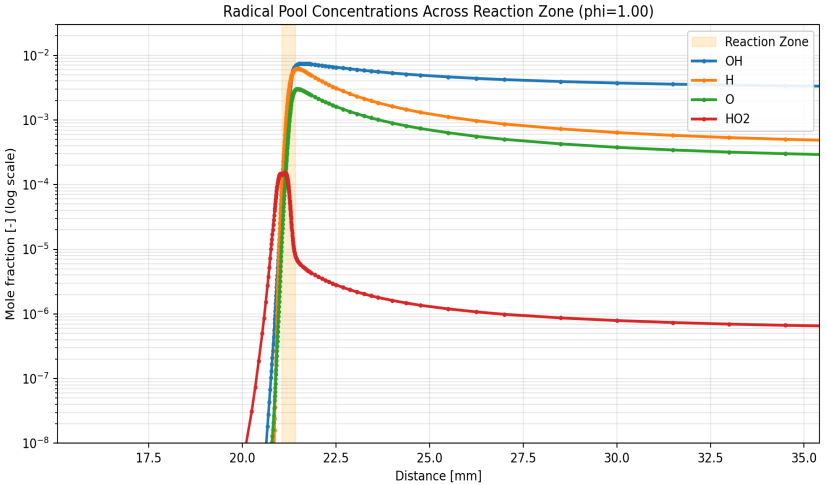


Three Characteristic Layers:

- **Preheat Zone (x < 21.06 mm):** Upstream conduction heats cold reactants from 300 K to ~1300 K without fuel consumption.
- **Reaction Zone (21.06 to 21.41 mm):** Marked by orange shading. Thickness δ_rxn = **352 μm** (only 0.35 mm). Volumetric heat release peaks at **4.50 GW/m³**.
- **Burned Zone (x > 21.41 mm):** Thermal expansion causes density drop (1.13 → 0.15 kg/m³) and velocity acceleration (0.38 → 2.87 m/s) to conserve mass flux.

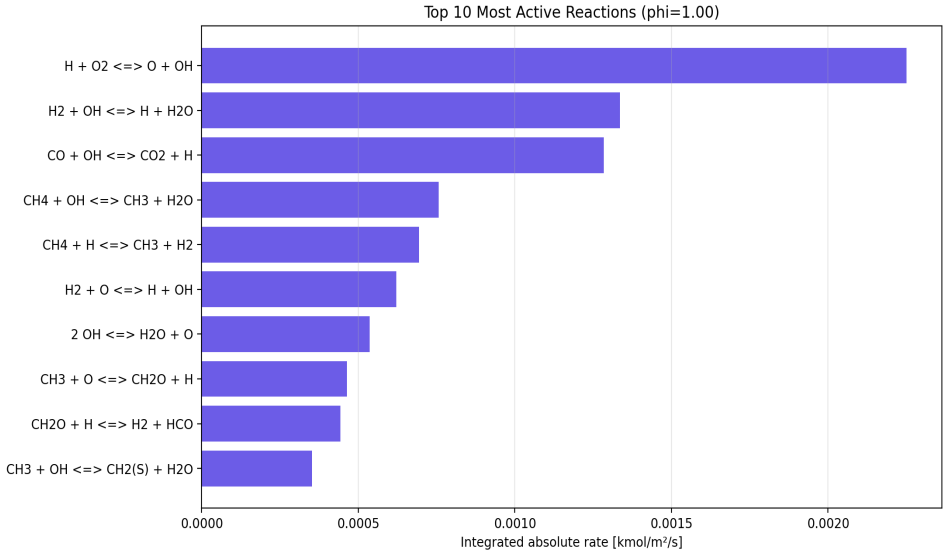
Results: Major Species & Radical Pool Evolution





Physical Insight: CH_4 and O_2 deplete sharply within the 350 µm reaction layer. Intermediate CO peaks inside the zone before oxidizing to CO_2. Radicals (OH, H, O, HO_2) exist at low mole fractions (10^{-3} to 10^{-6}) and peak strictly within the heat release zone to sustain chain branching.

Results: Elementary Chemical Kinetics Ranking



Dominant Elementary Reactions:

1. H + O2 ⇔ O + OH

Primary chain-branching reaction. Converts 1 radical into 2. Governs overall laminar flame speed.

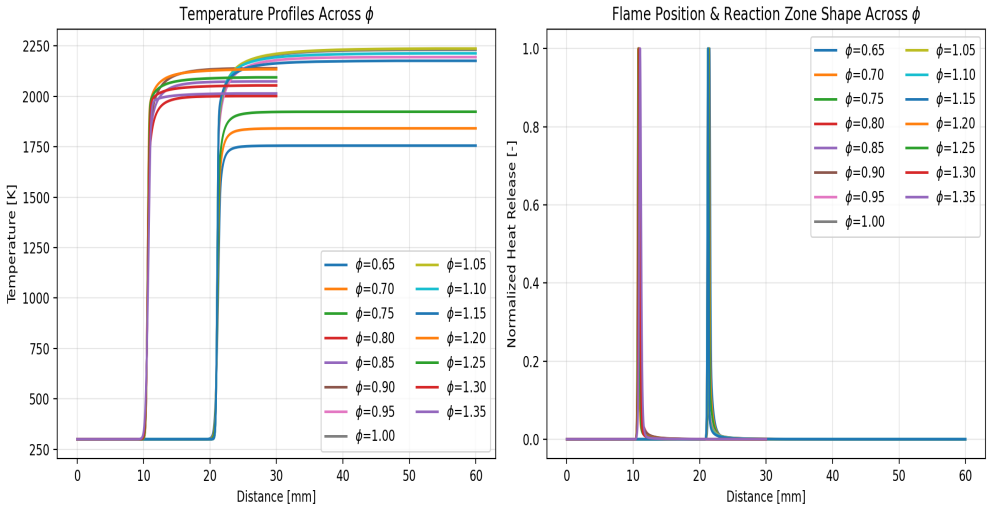
2. CO + OH ⇔ CO2 + H

Primary exothermic heat release reaction. Produces the bulk of thermal energy in hydrocarbon combustion.

3. OH + H2 ⇔ H2O + H

Rapid radical propagation producing stable water vapor.

Results: Flame Structure Across Equivalence Ratios



Cross-Phi Dynamics:

- Near stoichiometric (ϕ = 1.0 - 1.1), thermal gradients are steepest, resulting in the thinnest reaction layers ($\delta_T \approx 0.43$ mm).
- In lean (ϕ = 0.7) and rich (ϕ = 1.3) regimes, reaction layers broaden significantly ($\delta_T \approx 0.62$ - 0.66 mm) due to lower reaction rates.
- **Fundamental Law Demonstrated:** Thinner Reaction Layer ⇔ Higher Temperature Gradient ⇔ Faster Flame Speed.

Numerical Verification & Literature Validation

Validation Parameter	Literature Benchmark	Our Simulation Output	Agreement / Error
Laminar Flame Speed (S_L at $\phi = 1.00$)	38.0 – 39.0 cm/s	**38.09** cm/s	$\Delta < 0.2\%$ (Exact)
Peak Flame Speed ($\phi = 1.05$)	~ 38.0 – 39.5 cm/s	**38.76** cm/s	Exact Peak Match
Adiabatic Flame Temperature (T_{ad})	2225 – 2235 K	**2230.7** K	Thermodynamic Limit
Mass Flux Conservation ($\dot{m} = \rho u$)	$\rho_u u_u$ = constant	$0.430 \rightarrow 0.430$ kg/(m$^2 \cdot$ s)	**0.00% Error**

Summary & Conclusions

1. **Successful 1D Flame Modeling:** Freely-propagating laminar methane-air flames were successfully solved across ϕ = 0.65 – 1.35 using detailed GRI-Mech 3.0 chemistry.

2. **Peak Phenomena Identified:** Global peak flame speed occurs at ϕ = 1.05 (38.76 cm/s) and maximum temperature reaches 2237.1 K, driven by radical pool enrichment (H, OH).

3. **Spatial Layering Quantified:** The chemical reaction zone was proven to be sub-millimeter thin (δ_rxn $\approx$ 352 μm), preceded by a conductive preheat zone.

4. **High Numerical Integrity:** Adaptive mesh refinement (16 $\rightarrow$ 240 points) ensured zero artificial diffusion and exact agreement (<0.2%) with standard combustion literature.

Thank You

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